

TAK Server 'X-Forwarded-For' Logging Enablement

This process is for organizations that use compiled RPMs from tak.gov and do not maintain a local TAK build server.

TAK Server's <https://github.com/TAK-Product-Center/Server> default HTTP access log records only the address of the upstream device that connects to it as the 'remoteip' entity. When the TAK server sits behind any load balancer or reverse web proxy, such as a BIG-IP, every tomcat HTTP log entry shows only the proxy's address, even though the proxy might be sending the X-Forwarded-For header to TAK. The tomcat log format is unfortunately fixed in compiled Java code and cannot be manipulated through TAK's CoreConfig.xml or startup script options. This process modifies the single java class that defines the embedded Tomcat log format, thus enabling every entry to also record the 'X-Forwarded-For' value. This permits every request to be traced to its originating client within TAK's syslog. This process does not require a grable rebuild of takserver.war.

This process has been validated against TAK Server 5.5-RELEASE-82 and should work for subsequent releases, provided that the TAK developers do not change the embedded Tomcat architecture.

Non-TAK-prod Rhel server tools preparation *(used for patching the java class into the war)*

1. Install the required packages

```
dnf install -y java-21-openjdk-devel unzip binutils
```

Windows client tools preparation *(used for modifying and exporting a java class from the war)*

2. Download Recaf, a Java bytecode analyzer and disassembler: <https://github.com/Col-E/Recaf>
3. Download a Java Development Kit (required for Recaf): <https://www.oracle.com/java/technologies/downloads/>

Back up the current unmodified war

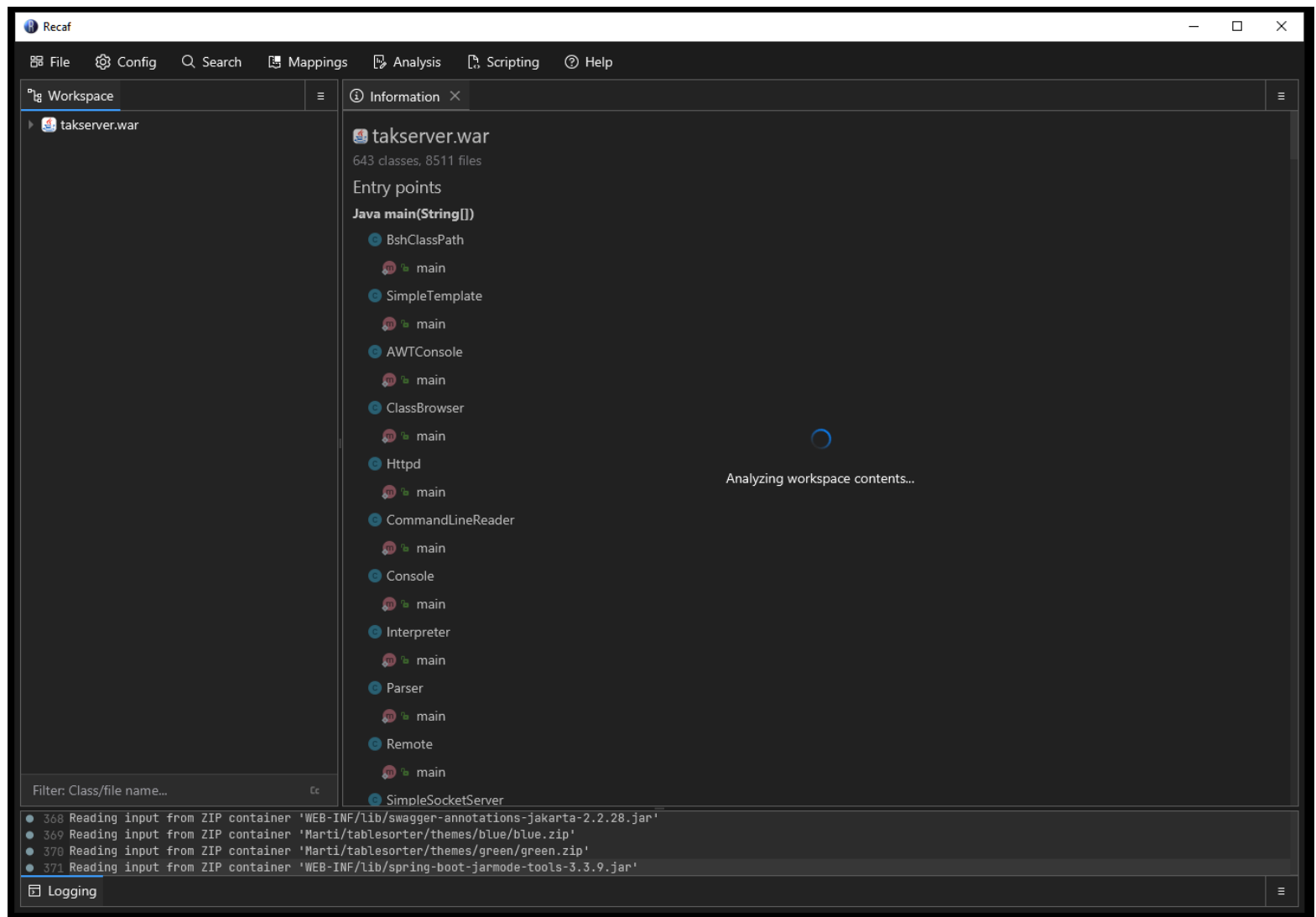
4. Copy the default war to a backup version

```
cp -a /opt/tak/takserver.war /opt/tak/takserver.war.backup
```

Process and modify the takserver.war and export a single modified class file

5. Copy takserver.war to the Windows client machine via scp
6. Open Recaf by double-clicking the Recaf jar file

7. Drag and drop the unmodified takserver.war into the Recaf window and wait until war analysis is complete



8. In the left workspace window, navigate to **Classes > tak > server > CustomizeEmbeddedTomcatContainer**

9. CustomizeEmbeddedTomcatContainer is the class that instructs Tomcat on how to log HTTP connection information. In the right panel find the line:

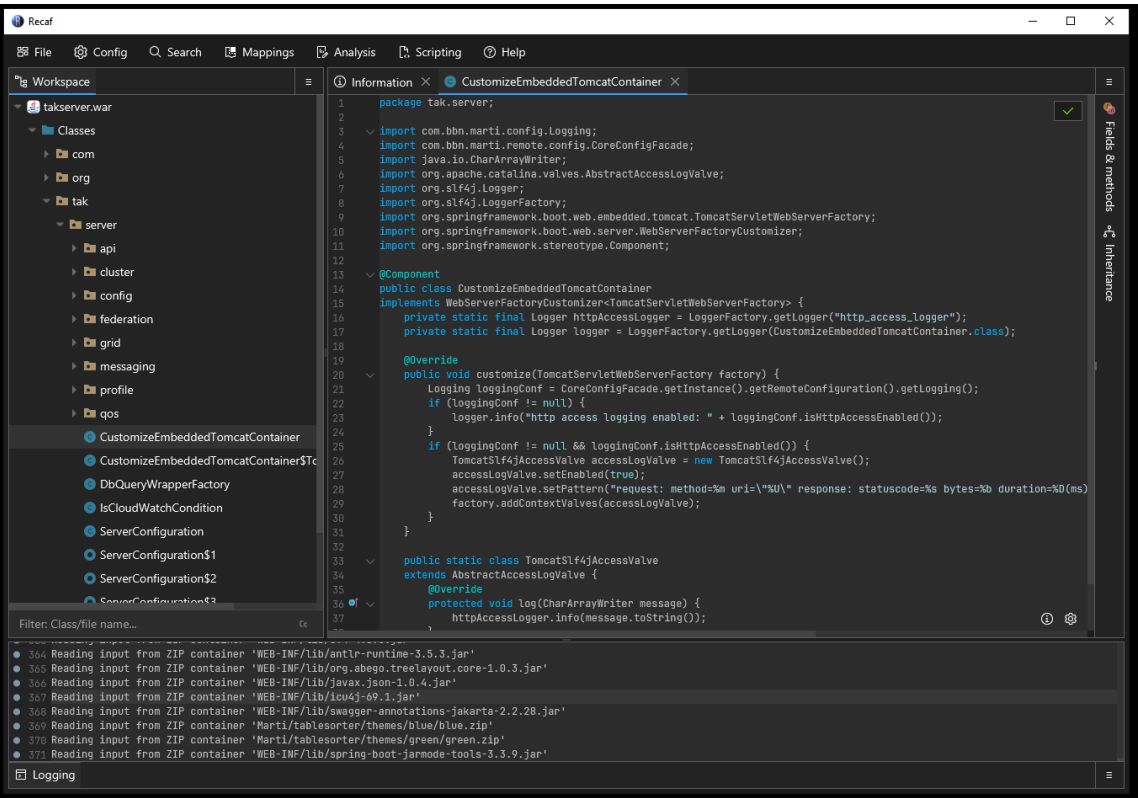
```
accessLogValve.setPattern("request: method=%m uri=\"%U\" response: statuscode=%s bytes=%b duration=%D(ms) client: remoteip=%a user=%{username}s useragent=\"%{User-Agent}i\");
```

10. And modify it to include the incoming X-Forwarded-For header via: **xff=%{X-Forwarded-For}i**

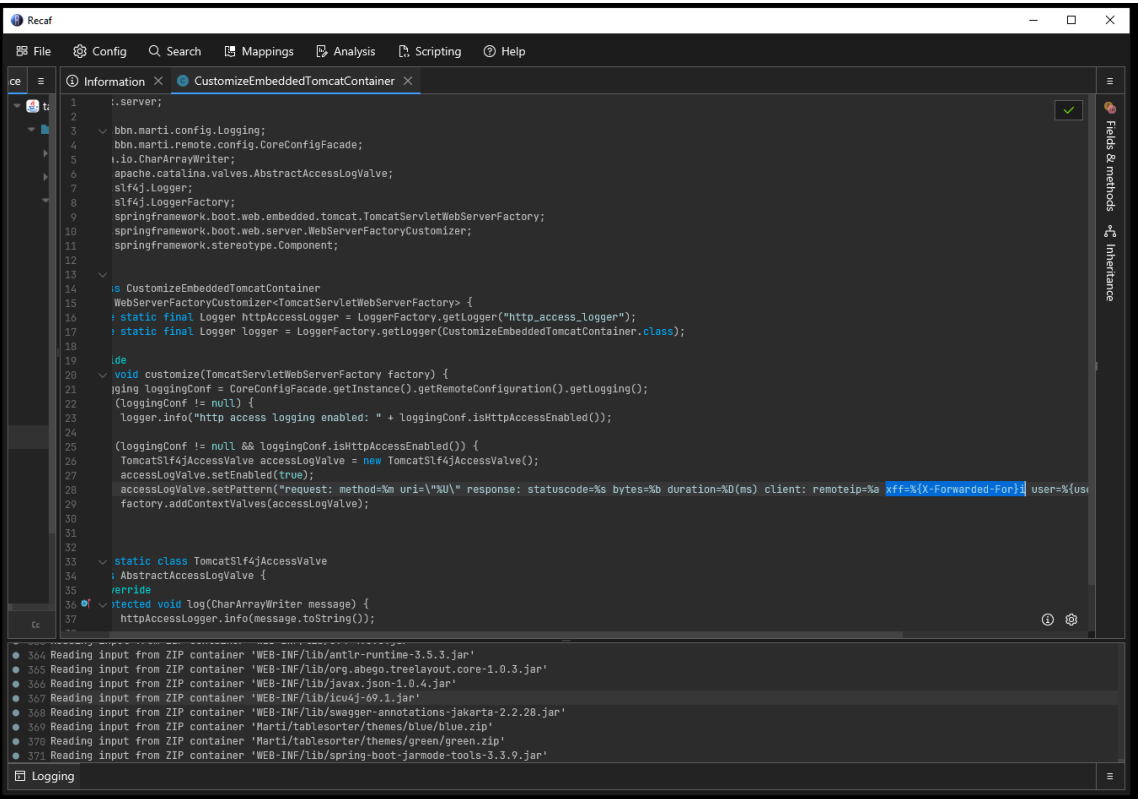
```
accessLogValve.setPattern("request: method=%m uri=\"%U\" response: statuscode=%s bytes=%b duration=%D(ms) client: remoteip=%a xff=%{X-Forwarded-For}i user=%{username}s useragent=\"%{User-Agent}i\");
```

11. Do not make any other changes. Press Control-S in Recaf to save the modified class file.

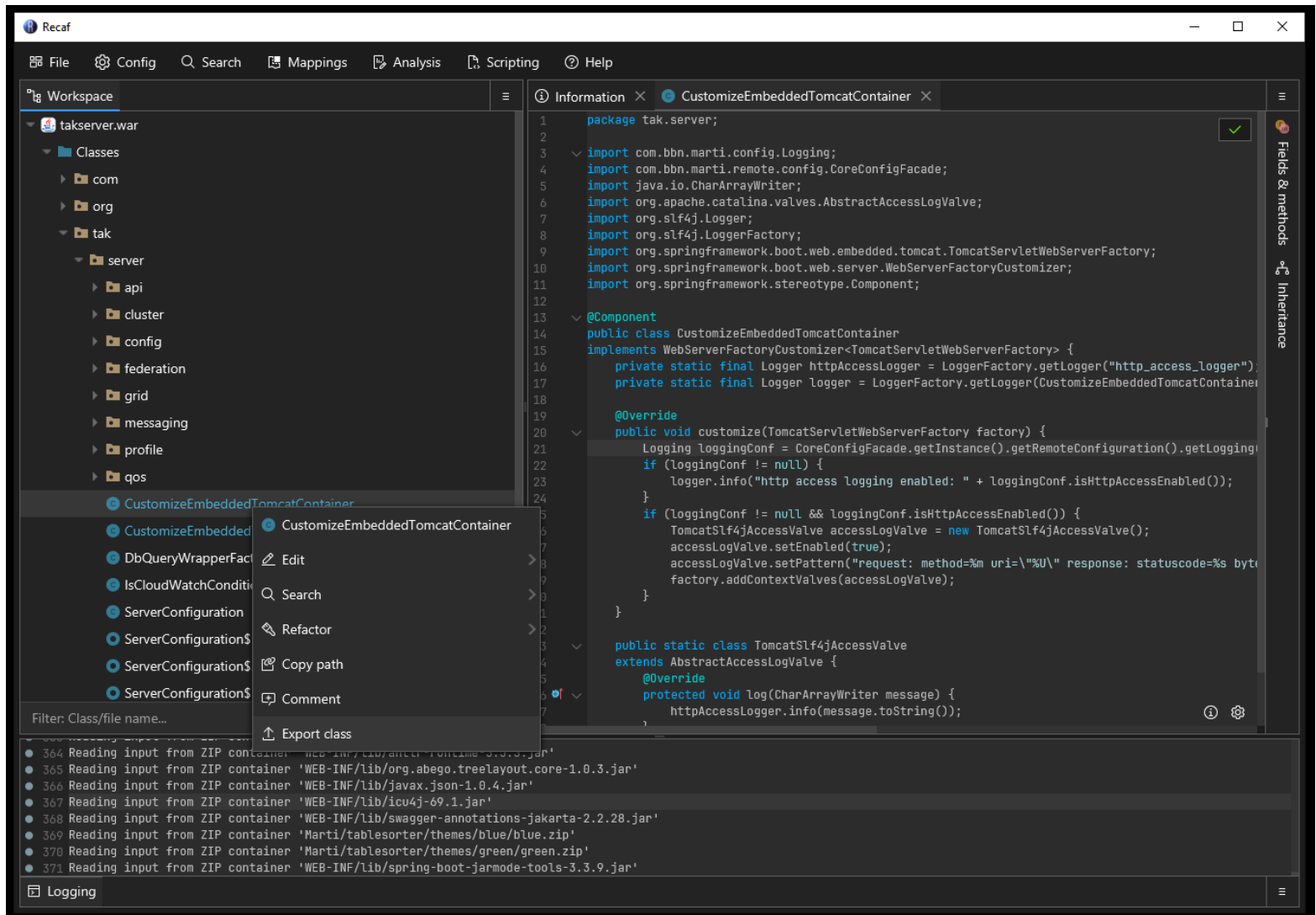
Original:



Modified:



12. Do not export the entire war using Recaf, this repackages the war in a layout the TAK Server cannot load. Right click on 'CustomizeEmbeddedTomcatContainer' and select "Export class". Recaf will ask you where to save the 'CustomizeEmbeddedTomcatContainer.class' file. Ensure you do not edit this file in windows notepad or any text editor or you could cause bytecode corruption of the exported class file rendering it useless.



13. Copy takserver.war and the modified class file to the Tak build server or a non-Tak RHEL server, into the same directory

Patch the modified class file into the war

14. Run these commands in the directory holding takserver.war and the modified class file
- jar places a file in the war at the same relative path it has on disk, so the class **must** be moved into a temporary **WEB-INF/classes/tak/server** folder which builds the necessary internal pathing in the war during the patch

```
mkdir -p WEB-INF/classes/tak/server
mv CustomizeEmbeddedTomcatContainer.class WEB-INF/classes/tak/server/
jar uf takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class
rm -rf WEB-INF
```

15. Verify the patch – The output **must be 1** before proceeding further

```
unzip -p takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class | strings | grep -c 'xff='
```

Install the patched war

16. On the TAK server, stop the service *(ensure to wait until the server has completely shutdown)*

```
systemctl stop takserver
```

17. From the Tak build server or a non-Tak RHEL server, copy the patched takserver.war to the TAK server at /opt/tak/takserver.war

```
scp takserver.war root@<tak-server>:/opt/tak/takserver.war
```

18. On the TAK server, set ownership and start the service

```
chown tak:tak /opt/tak/takserver.war  
systemctl start takserver
```

Logging Verification

19. Review the access log

```
tail -f /opt/tak/logs/takserver-api-access.log
```

Expected entry:

```
... client: remoteip=<BIG-IP address> xff=<client address> user=... useragent="..."
```

Rollback

20. Restore the backup and restart TAK Server *(ensure to wait until the server has completely shutdown)*

```
systemctl stop takserver  
cp -a /opt/tak/takserver.war.backup /opt/tak/takserver.war  
systemctl start takserver
```